



An Intelligent Decision-Support Framework for Sustainable Waste-to-Energy Planning and Smart Energy Infrastructure Under Uncertainty

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Abstract: Sustainable waste-to-energy (WtE) planning requires coordinated decisions on feedstock valorization, conversion technology, plant siting, operating strategy, and the utilization of recovered energy and by-products. These decisions are complicated by conflicting economic, environmental, technical, and social criteria, together with uncertainty and hesitation in operational and stakeholder information. This study develops an intelligent decision-support framework for integrated WtE planning and smart-energy infrastructure under uncertain information. Six sine-based similarity measures are extended from intuitionistic fuzzy sets to q -rung orthopair and T -spherical fuzzy environments, allowing support, opposition, hesitation, and stakeholder abstention to be represented within a common decision structure. The framework integrates data acquisition from supervisory control and data acquisition systems, smart meters, municipal information, and expert assessments with fuzzification, entropy-based weighting, similarity-based ranking, the Aggregate Separation Index (ASI), and Monte Carlo rank-stability analysis. Its performance is examined through four illustrative energy-planning cases involving waste-valorization pathway selection, energy-facility siting, conversion-route selection, operating-strategy selection for a biogas combined heat and power (CHP) unit, and industrial-symbiosis partner selection. Anaerobic digestion is consistently identified as the preferred valorization pathway, while biogas CHP is the leading conversion route. Under entropy-derived weights, Baseload is the preferred CHP operating strategy, although Hybrid-adaptive operation becomes more attractive when revenue and grid-support priorities receive greater emphasis. Under the reference setting, Monte Carlo analysis gives rank-1 probabilities of 0.948 for pathway selection and 0.999 for operating-strategy selection. By contrast, the two leading industrial-symbiosis partners differ by only 0.008, and the nominal leader retains first place with a probability of 0.632, so the decision is classified as contested. A machine-learning benchmark further shows that the similarity-based matcher remains within 0.6–3.9 percentage points of the best crisp prototype benchmark while retaining explicit information on similarity, hesitation, and aggregate separation. The results demonstrate how uncertainty-aware decision support can connect waste valorization, energy recovery, conversion technology, grid-responsive operation, and industrial symbiosis within an integrated framework for sustainable urban energy planning.

Keywords: Waste-to-energy; Sustainable energy systems; Smart energy infrastructure; Energy recovery; Biogas combined heat and power; Decision support; Uncertainty analysis; Circular economy

1 Introduction

Municipal solid waste (MSW) is increasingly being considered not only as an environmental burden but also as a potential resource for energy recovery and circular resource utilization. Global MSW generation was estimated at approximately 2.0 billion tonnes in 2016 and is projected to reach about 3.4 billion tonnes by 2050 [1]. For waste fractions that cannot be economically reused or recycled, waste-to-energy (WtE) technologies can contribute to the recovery of electricity, heat, fuels, and secondary materials while reducing the amount of waste requiring final disposal [2–4]. Their role is particularly relevant in urban energy systems, where local waste streams can be connected with district heating, distributed generation, industrial heat demand, and other components of integrated energy infrastructure.

The planning of WtE systems, however, involves a sequence of closely connected decisions rather than a single technology-selection problem. Decision makers may need to determine which valorization pathway is most appropriate, where a facility should be located, which conversion technology should be adopted, how the plant should respond to variations in energy prices and demand, and how recovered heat or by-products can be integrated with surrounding industries [5, 6]. These decisions involve economic, environmental, technical, operational, and social criteria that may conflict with one another. A technology with high energy-recovery potential may require substantial capital investment, while a lower-cost option may provide less favorable carbon-reduction or grid-support performance. Similarly, a technically suitable facility site may face land-use constraints, environmental sensitivity, infrastructure limitations, or low public acceptance.

Multi-criteria decision-making (MCDM) has therefore been widely applied to waste-management and WtE planning [7, 8]. Nevertheless, practical energy-planning problems often involve incomplete, qualitative, uncertain, or conflicting information. Fuzzy-set-based approaches provide a natural way to represent such uncertainty [9–11], particularly when technical indicators must be evaluated together with expert judgments concerning social acceptance, technology maturity, contractual reliability, or stakeholder preference. From a broader planning perspective, operations-research studies have also shown that waste-management decisions involve interconnected strategic and tactical issues [12], while smart-energy research has emphasized the coordinated planning of electricity, heat, transport, and distributed resources [13].

Data-driven methods provide an additional source of decision information. Artificial intelligence and machine-learning techniques have been increasingly applied to waste-generation forecasting, process prediction, classification, and operational assessment [14, 15]. Their predictive capability can support energy-management decisions, but purely data-driven models may provide limited explanation of why one infrastructure or operating alternative is preferred over another. This limitation is important in strategic WtE planning, where decisions must often be justified in terms of energy recovery, cost, carbon performance, operational risk, and stakeholder acceptance.

Classical MCDM methods such as technique for order preference by similarity to an ideal solution remain useful when evaluation information can be represented through well-defined numerical values [16]. However, conventional crisp formulations become less suitable when support and opposition toward the same alternative are both strong or when stakeholders explicitly express hesitation or neutrality. Pythagorean, Fermatean, and q -rung orthopair fuzzy sets extend the admissible decision space beyond conventional intuitionistic fuzzy sets [17–19], while picture fuzzy and T -spherical fuzzy models additionally retain neutral or abstention information [20, 21]. These extensions are particularly relevant to energy-infrastructure planning because technical, economic, environmental, and stakeholder assessments may exhibit different forms and degrees of uncertainty.

Despite these advances, two limitations remain important. First, many existing decision-support studies focus primarily on the final ranking while giving less attention to the robustness and separation of that ranking. In long-lived WtE infrastructure decisions, a very small difference between the two highest-ranked alternatives should not be interpreted in the same way as a large and stable ranking gap. Decision-support systems should therefore provide not only an ordered list of alternatives but also information on ranking stability, sensitivity to criterion weights, and the extent to which the leading alternative is separated from its competitors.

Second, different stages of WtE planning are still frequently treated as separate problems. Technology selection, facility siting, waste-management strategy, plant operation, and downstream resource utilization are often analyzed independently, even though they are strongly interrelated. Waste composition influences conversion technology, conversion technology determines the form and quantity of recoverable energy, plant location affects grid connection and heat-offtake opportunities, and operating strategy influences revenue, emissions, and flexibility. A more integrated decision architecture is therefore needed to connect these stages while retaining uncertainty and interpretability.

Against this background, this study develops an intelligent decision-support framework for sustainable WtE planning and smart-energy infrastructure under uncertain information. The framework extends six sine-based similarity measures from intuitionistic fuzzy information to q -rung orthopair and T -spherical fuzzy environments and integrates them with entropy-based criterion weighting, similarity-based ranking, the Aggregate Separation Index (ASI), and Monte Carlo rank-stability analysis. The framework combines operational and municipal information with uncertain expert assessments and is designed to identify both well-supported recommendations and situations in which the available evidence is insufficient to justify a unique decision.

The main contributions of the study are fourfold. First, the mathematical properties of the extended similarity measures are examined for q -rung orthopair and T -spherical fuzzy numbers, including admissibility, monotonicity, metric behavior, and entropy-related properties. Second, a five-layer decision-support architecture is developed to connect data acquisition, fuzzification, criterion weighting, similarity-based reasoning, and decision-separation assessment within a common framework. Third, the framework is demonstrated through four WtE planning cases involving five interrelated decisions: valorization-pathway selection, facility siting, conversion-route selection under stakeholder abstention, operating-strategy selection for a biogas combined heat and power (CHP) unit, and industrial-

symbiosis partner selection. These cases cover different stages of sustainable energy planning, from resource conversion and plant configuration to grid-responsive operation and the utilization of recovered heat and by-products. Fourth, the robustness of the decision layer is evaluated through sensitivity analysis, Monte Carlo perturbation, and machine-learning benchmarks to determine whether the proposed similarity-based representation can provide interpretable decisions without a substantial loss of classification performance.

The remainder of the paper is organized as follows. Section 2 introduces the fuzzy-information concepts required for the proposed framework. Section 3 presents the extended sine-based similarity measures and examines their mathematical properties. Section 4 describes the proposed decision-support methodology and its integration with energy-management information. Section 5 applies the framework to four WtE planning cases. Section 6 evaluates the machine-learning integration and robustness of the similarity-based decision layer. Section 7 discusses the implications for sustainable energy planning and the main limitations of the study. Section 8 summarizes the principal conclusions and directions for future research.

2 Preliminaries

This section introduces the two fuzzy-information structures required for the proposed decision framework. They are employed because several evaluations encountered in sustainable WtE planning cannot be represented adequately by conventional intuitionistic fuzzy numbers.

2.1 q -Rung Orthopair Fuzzy Number

A q -rung orthopair fuzzy number (q -ROFN) [19] is represented by a pair $\alpha = (\mu, \nu)$, where μ and ν denote the membership and non-membership degrees, respectively, subject to $\mu^q + \nu^q \leq 1, q \geq 1$.

The associated hesitation degree is $\pi = (1 - \mu^q - \nu^q)^{1/q}$.

The value of q determines the admissible information space. In particular, intuitionistic, Pythagorean, and Fermatean fuzzy numbers can be regarded as particular cases of the q -rung orthopair family for the corresponding values of q . This flexibility is useful in energy-planning problems in which support for and opposition to the same alternative may both be relatively high.

For example, when $\mu + \nu > 1$, an intuitionistic fuzzy representation is no longer admissible. A q -rung orthopair representation can nevertheless remain feasible provided that $\mu^q + \nu^q \leq 1$.

This property is relevant to the evaluation of energy-conversion technologies, infrastructure locations, and operating strategies for which technical, environmental, economic, and stakeholder assessments may generate relatively strong but conflicting judgments.

2.2 T -Spherical Fuzzy Number

A T -spherical fuzzy number (T -SFN) [21] is represented by an ordered triplet $\beta = (\mu, \eta, \nu)$, where μ , η , and ν denote the membership, neutral or abstention, and non-membership degrees, respectively. These components satisfy $\mu^q + \eta^q + \nu^q \leq 1$.

The associated refusal degree may therefore be written as $\pi = (1 - \mu^q - \eta^q - \nu^q)^{1/q}$.

The explicit neutral component η allows stakeholder abstention to be retained rather than forcing an uncertain assessment into either support or opposition. Picture and spherical fuzzy numbers can be treated as special cases under the corresponding restrictions on the rung parameter.

In the present framework, q -ROFNs are used when the decision information contains conflicting support and opposition without an explicit abstention component, whereas T -SFNs are adopted when stakeholder neutrality or abstention must also be represented. This distinction is particularly relevant to sustainable energy planning because technical performance indicators often coexist with uncertain social, regulatory, environmental, and commercial judgments.

3 Sine-Based Similarity Measures for Hybrid Fuzzy Numbers

Similarity and distance measures provide an important basis for comparing uncertain information in fuzzy decision-making. Recent and classical studies have developed cosine-based similarity measures for intuitionistic fuzzy sets and demonstrated their usefulness in decision analysis [22, 23], while distance-based formulations provide a complementary means of quantifying separation between intuitionistic fuzzy information [24].

Building upon these developments, the present framework extends the six sine-based similarity measures introduced in the work of Khan et al. [25] from intuitionistic fuzzy information to q -rung orthopair and T -spherical fuzzy environments. The purpose of this extension is to establish a common similarity structure capable of handling different forms of uncertainty encountered in WtE and smart-energy infrastructure planning.

3.1 Coordinate Representation

The measures proposed in the work of Khan et al. [25] depend on the absolute differences between the information coordinates of two intuitionistic fuzzy numbers. To extend this principle to q -ROFNs and T -SFNs, the original membership components are transformed into coordinates that preserve the geometry of the corresponding fuzzy domain.

For a q -ROFN:

$$\alpha = (\mu, \nu) \quad (1)$$

the transformed coordinate representation is based on:

$$\phi_q(\alpha) = (\mu^q, \nu^q) \quad (2)$$

For a T -SFN:

$$\beta = (\mu, \eta, \nu) \quad (3)$$

the corresponding transformed representation is:

$$\phi_q(\beta) = (\mu^q, \eta^q, \nu^q) \quad (4)$$

For two fuzzy numbers A and B , let the absolute difference in the l -th transformed coordinate be:

$$\delta_l = |\phi_l(A) - \phi_l(B)| \quad (5)$$

The total coordinate difference is defined as:

$$\Delta = \sum_l \delta_l \quad (6)$$

and the generalized coordinate-distance term is:

$$M_p = \left(\frac{1}{2} \sum_l \delta_l^p \right)^{1/p}, \quad p \geq 1 \quad (7)$$

Using these transformed coordinate differences, the six extended sine-based similarity kernels are defined as:

$$\begin{aligned} S_1 &= 1 - \sin\left(\frac{\pi\Delta}{1+\Delta}\right) \\ S_2 &= 1 - \sin\left(\frac{\pi\Delta^2}{8}\right) \\ S_3 &= 1 - \sin\left[\frac{\pi}{8}\Delta\left(1 + \min_l \delta_l\right)\right] \\ S_4 &= \left[1 - \sin\left(\frac{\pi}{2}M_p\right)\right]^2 \\ S_5 &= 1 - \frac{2\sin\left(\frac{\pi}{2}M_p\right)}{1 + \sin\left(\frac{\pi}{2}M_p\right)} \\ S_6 &= 1 - \sin\left(\frac{\pi\Delta}{4}\right) \end{aligned} \quad (8)$$

For a decision problem involving n criteria with normalized criterion weights w_j ,

$$w_j \geq 0, \quad \sum_{j=1}^n w_j = 1 \quad (9)$$

the overall similarity between two alternatives, or between an alternative and an ideal profile, is obtained by weighted aggregation across the criterion-level similarities:

$$S(A, B) = \sum_{j=1}^n w_j s_k(A_j, B_j), \quad k = 1, \dots, 6 \quad (10)$$

Here, A_j and B_j denote the fuzzy representations of A and B under criterion C_j , respectively. This formulation allows alternatives represented in different admissible fuzzy domains to be compared with positive or negative ideal energy-performance profiles within a common similarity framework.

3.2 Mathematical Properties

For a similarity measure to provide reliable decision support, it should satisfy several basic requirements.

For two fuzzy numbers A and B , the similarity measure should satisfy boundedness, $0 \leq S(A, B) \leq 1$, identity, $S(A, B) = 1 \iff A = B$, and symmetry, $S(A, B) = S(B, A)$.

A further desirable property is monotonicity with increasing coordinate separation. If B lies between A and C , the similarity should satisfy an ordering such as $S(A, C) \leq S(A, B)$, because a larger disagreement between two information profiles should not result in a larger similarity.

Theorem 1

For the admissible q -rung orthopair and T -spherical fuzzy domains considered in this study, the extended measures satisfy the boundedness, identity, and symmetry requirements. Measures S_2 – S_6 satisfy the required monotonicity condition throughout their admissible domains, whereas S_1 satisfies this condition only over the region in which its sine argument remains on the appropriate monotonic branch.

Proof

Boundedness follows from the range of the corresponding sine function over the admissible coordinate interval. Accordingly, $0 \leq S_i(A, B) \leq 1$.

Identity follows because the transformed coordinate distance becomes zero only when the corresponding fuzzy-information coordinates coincide: $\Delta_k = 0 \forall k \iff A = B$.

Symmetry follows directly from the use of absolute differences: $|x_k - y_k| = |y_k - x_k|$.

Therefore, $S_i(A, B) = S_i(B, A)$.

For the monotonicity property, increasing disagreement between the transformed coordinates increases the corresponding distance measure. For S_2 – S_6 , the associated kernel remains non-increasing throughout the admissible interval. Consequently, $\Delta_1 \leq \Delta_2 \implies S_i(\Delta_1) \geq S_i(\Delta_2)$.

The same relation holds for S_1 only within the range over which its sine kernel remains monotonic.

Remark 1

The restricted monotonicity of S_1 is important for practical decision making. When its sine argument moves beyond the monotonic interval, the similarity may increase even though the coordinate distance continues to increase. As shown in Figure 1a, S_1 reaches its minimum at $\Delta = 1$ and increases again when $\Delta > 1$, whereas the remaining admissible kernels decrease monotonically over the considered coordinate-distance range. Thus, it is possible to have $\Delta_1 < \Delta_2$ while $S_1(\Delta_1) < S_1(\Delta_2)$, which contradicts the expected interpretation of similarity. Accordingly, S_1 is retained only as a comparison measure and is not recommended as the default kernel for unrestricted ideal-solution evaluations.

This limitation also has a direct energy-planning interpretation. If an alternative moves farther away from the ideal profile in terms of energy recovery, carbon intensity, grid flexibility, technology readiness, or other sustainability criteria, the similarity measure should not assign it a higher similarity solely because of the mathematical behavior of the kernel.

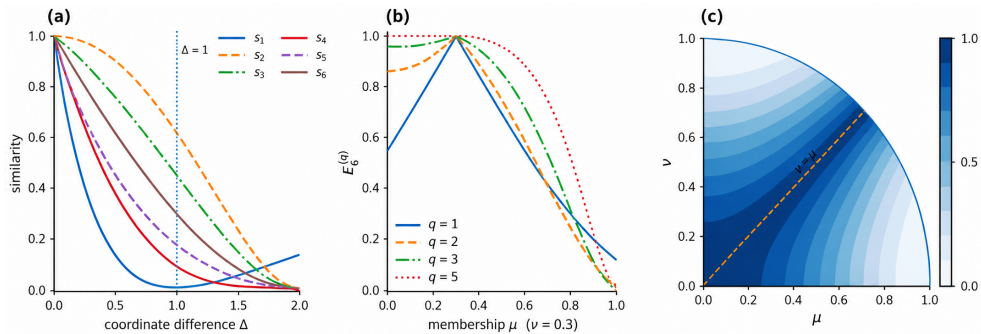


Figure 1. Mathematical behavior of the extended sine-based measures: (a) six similarity kernels as functions of coordinate distance at $p = 2$, showing the loss of monotonicity of S_1 beyond $\Delta = 1$; (b) sine entropy $E_6^{(q)}$ for different rung values at $\nu = 0.3$; (c) $E_6^{(2)}(\mu, \nu)$ over the Pythagorean fuzzy domain

3.3 Metric Properties

The induced distance associated with a similarity measure can generally be expressed in the form $d(A, B) = f(\Delta(A, B))$, where $f(\cdot)$ transforms the coordinate difference into a distance.

A valid metric should satisfy $d(A, B) \geq 0$, $d(A, B) = 0 \iff A = B$, $d(A, B) = d(B, A)$, and the triangle inequality $d(A, C) \leq d(A, B) + d(B, C)$.

Theorem 2

As summarized in Table 1, the induced distances associated with S_1 , S_4 , S_5 , and S_6 satisfy the triangle inequality, whereas this property does not hold generally for S_2 and S_3 ; the corresponding counterexamples follow from the construction discussed in the work of Khan et al. [25].

Table 1. Mathematical properties of the extended sine-based similarity measures

Measure	(S1)–(S4)	Metric	(P1)	(P2)	Entropy	Parameter
S_1	$\Delta \leq 1$ only	Yes	No	Yes	No	None
S_2	Yes	No	Yes	Yes	Yes	None
S_3	Yes	No	q -ROFN only	Yes	q -ROFN only	None
S_4	Yes	Yes	Yes	Yes	Yes	$p \geq 1$
S_5	Yes	Yes	Yes	Yes	Yes	$p \geq 1$
S_6	Yes	Yes	Yes	Yes	Yes	None

Note: q -ROFN = q -rung orthopair fuzzy number.

Table 1 shows that S_4 , S_5 , and S_6 satisfy the complete set of similarity, metric, and entropy-related properties considered in this study. S_1 retains metric properties but has restricted monotonicity for $\Delta > 1$, whereas S_2 and S_3 do not generally induce metrics. Accordingly, S_6 is used as the principal parameter-free measure in the subsequent analysis, while the remaining kernels are retained for comparison and sensitivity testing.

The result follows from the metric properties of the transformed coordinate representation together with the monotonic, concave, and subadditive characteristics of the corresponding sine-based transformation.

For the admissible measures, $d(A, C) \leq d(A, B) + d(B, C)$.

The counterexample presented in the work of Khan et al. [25] can be transferred to the hybrid fuzzy environment through the q -rung coordinate transformation.

These properties are relevant to the robustness analysis conducted later in the study because the proposed framework repeatedly evaluates how rankings change when the energy-performance ratings are subjected to small perturbations.

3.4 Entropy-Based Criterion Weighting

The same similarity family is used to derive an entropy measure for criterion weighting. For criterion C_j , let its entropy be denoted by E_j . The corresponding information divergence and normalized entropy weight are defined as:

$$D_j = 1 - E_j, \quad w_j = \frac{D_j}{\sum_{k=1}^n D_k} \quad (11)$$

The resulting weights satisfy:

$$w_j \geq 0, \quad \sum_{j=1}^n w_j = 1 \quad (12)$$

The entropy-weighting mechanism assigns a larger weight to criteria that contain greater discriminatory information among the alternatives. Conversely, criteria for which the alternatives exhibit relatively similar evaluations receive smaller weights.

Corollary 1

For the similarity measures identified in Table 1, the corresponding entropy formulation satisfies the required entropy properties over the specified q -rung orthopair domain.

The entropy-weighting mechanism assigns a larger weight to criteria that contain greater discriminatory information among the alternatives. Conversely, criteria for which the alternatives exhibit relatively similar evaluations receive smaller weights.

This interpretation is particularly appropriate for the energy-planning applications considered in this study. For example, if candidate conversion routes differ substantially in energy-recovery efficiency, carbon intensity, grid flexibility, or technology readiness, these criteria provide stronger discriminatory information and can therefore receive larger entropy-derived weights.

Conversely, if all alternatives show similar performance with respect to a particular criterion, that criterion contributes less information to the ranking.

The rung parameter q affects the geometry of the fuzzy-information space and therefore influences both similarity and entropy calculations. Likewise, the norm parameter p used in the distance-based kernels controls the relative importance of average and maximum coordinate disagreement. As p increases, greater emphasis is placed on the coordinate exhibiting the largest mismatch.

The influence of the rung parameter on the sine entropy is illustrated in Figure 1b. For a fixed non-membership degree, changes in q modify the shape of the entropy profile while preserving maximum uncertainty near balanced membership and non-membership. Figure 1c further shows the distribution of $E_6^{(2)}(\mu, \nu)$ over the Pythagorean domain, with the maximum entropy occurring along $\mu = \nu$.

Table 1 summarizes the principal properties of the six extended similarity measures. Measures satisfying the complete set of required properties are preferred in the subsequent decision analysis, whereas measures with restricted monotonicity or metric behavior are retained for sensitivity and benchmark comparisons.

4 Proposed Intelligent Decision-Support Methodology

The proposed methodology integrates heterogeneous energy-system information, fuzzy uncertainty representation, criterion weighting, similarity-based reasoning, and decision-robustness assessment within a five-layer decision-support architecture. It is designed for sustainable WtE planning problems in which measured operational information must be considered together with uncertain expert and stakeholder assessments.

The overall architecture and its interaction with the energy management system (EMS) are illustrated in Figure 2. The five layers comprise: (1) acquisition of operational, municipal, and stakeholder information; (2) normalization and fuzzy representation; (3) entropy-based and, where required, blended criterion weighting; (4) similarity-based reasoning against ideal or reference profiles; and (5) decision-support outputs consisting of rankings, separation information, and robustness indicators.

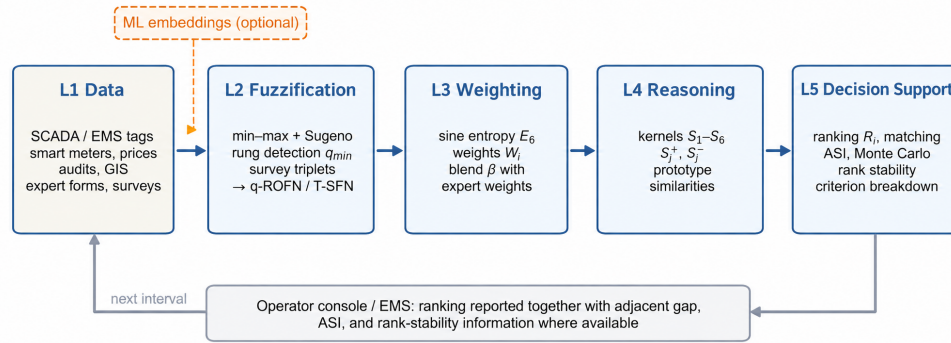


Figure 2. Five-layer intelligent decision-support architecture and its interaction with the EMS

Note: ML = machine learning; SCADA = supervisory control and data acquisition; GIS = geographic information system; q -ROFN = q -rung orthopair fuzzy number; T -SFN = T -spherical fuzzy number; ASI = Aggregate Separation Index; EMS = energy management system.

A distinguishing feature of the framework is that the same architecture can support decisions occurring at different planning horizons. Strategic decisions, such as waste-valorization pathway selection, conversion-route selection, and facility siting, may remain unchanged for extended periods. By contrast, operational decisions, such as the preferred operating strategy of a biogas CHP unit, may be reconsidered when energy prices, grid requirements, heat demand, forecast uncertainty, or other operating conditions change.

The framework is therefore not intended as a replacement for physical process models, plant-level control systems, or detailed energy-system optimization. Rather, it provides a supervisory decision layer in which heterogeneous information can be translated into comparable uncertain representations and evaluated in a transparent manner. When the input matrix changes because of updated data, criteria, or decision priorities, the weighting, similarity, and robustness calculations can be repeated to obtain an updated decision-support output.

4.1 Layer 1: Energy-System Data Acquisition

The first layer collects the information required to characterize the candidate alternatives and evaluation criteria. Depending on the decision context, the input data may originate from:

- supervisory control and data acquisition (SCADA) systems;
- smart meters;
- municipal waste statistics;
- waste-composition analyses;
- electricity and heat demand records;

- market and tariff information;
- grid-operating data;
- environmental indicators;
- technology-readiness assessments;
- expert judgments; and
- stakeholder evaluations.

These data sources may differ substantially in scale, uncertainty, update frequency, and reliability. Measured variables such as waste generation, moisture content, electricity output, heat recovery, or distance can often be expressed numerically, whereas social acceptance, contractual reliability, technology maturity, and stakeholder preference may require linguistic or fuzzy assessment.

The framework therefore separates data acquisition from uncertainty representation so that both measured and judgment-based information can be integrated without forcing all criteria into a single deterministic format.

4.2 Layer 2: Normalization and Fuzzy Representation

The second layer transforms heterogeneous numerical and judgment-based information into a comparable fuzzy representation. Let the original decision matrix be:

$$X = [x_{ij}]_{m \times n} \quad (13)$$

where, m denotes the number of alternatives and n the number of evaluation criteria.

For a numerical criterion C_j , the observed value x_{ij} is first normalized using min–max scaling:

$$\mu_{ij} = \frac{x_{ij} - \min_i \{x_{ij}\}}{\max_i \{x_{ij}\} - \min_i \{x_{ij}\} + \varepsilon} \quad (14)$$

where, ε is a small positive constant introduced to avoid division by zero.

The corresponding non-membership degree is obtained using the Sugeno complement [26]:

$$\nu_{ij} = \frac{1 - \mu_{ij}}{1 + \lambda \mu_{ij}}, \lambda \geq 0 \quad (15)$$

For cost-type criteria, the preference direction is reversed before subsequent aggregation so that larger membership values consistently represent more desirable performance.

The resulting pair $\alpha_{ij} = (\mu_{ij}, \nu_{ij})$ is then checked against the admissibility condition of the required fuzzy domain. For a q -rung orthopair representation, $\mu_{ij}^q + \nu_{ij}^q \leq 1$. If explicit neutrality or abstention is present, the evaluation is represented instead as a T -spherical fuzzy number, $\alpha_{ij} = (\mu_{ij}, \eta_{ij}, \nu_{ij})$, subject to $\mu_{ij}^q + \eta_{ij}^q + \nu_{ij}^q \leq 1$.

Related fuzzification strategies have been used to transform normalized numerical information into intuitionistic fuzzy representations for subsequent similarity-based analysis [27]. In the present framework, the selected fuzzy family is determined by the structure of the underlying information rather than imposed uniformly across all decision problems.

4.3 Layer 3: Entropy-Based Criterion Weighting

The third layer determines criterion importance from the discriminatory information contained in the fuzzy decision matrix. For criterion C_j , its average entropy is calculated as:

$$E_j = \frac{1}{m} \sum_{i=1}^m E_6^q(\alpha_{ij}) \quad (16)$$

where, E_6^q is the sine-based entropy defined in Section 3.

The corresponding entropy weight is:

$$w_j = \frac{1 - E_j}{\sum_{l=1}^n (1 - E_l)} \quad (17)$$

with

$$w_j \geq 0, \sum_{j=1}^n w_j = 1 \quad (18)$$

A criterion receives a larger weight when it provides greater discriminatory information among the candidate alternatives, whereas criteria with more homogeneous evaluations receive smaller weights.

The entropy-derived weights represent objective information contained in the decision matrix rather than fixed policy preferences. When subjective priorities are also available, the entropy-based weight vector $w^{(E)}$ can be combined with an expert-derived vector $w^{(S)}$ using:

$$w(\beta) = (1 - \beta)w^{(E)} + \beta w^{(S)}, \quad 0 \leq \beta \leq 1 \quad (19)$$

Here, $\beta = 0$ corresponds to fully entropy-derived weights, while increasing β progressively increases the influence of expert priorities. This blended weighting scheme is used later to examine how changes in revenue and grid-support priorities affect the preferred CHP operating strategy.

4.4 Layer 4: Similarity-Based Ranking

After criterion weighting, each alternative is compared with positive and negative ideal fuzzy profiles.

For q -rung orthopair information, the positive and negative ideal solutions for criterion C_j are defined as:

$$\alpha_j^+ = \left(\max_i \mu_{ij}, \min_i \nu_{ij} \right), \quad \alpha_j^- = \left(\min_i \mu_{ij}, \max_i \nu_{ij} \right) \quad (20)$$

For alternative A_i , its weighted similarities to the positive and negative ideal solutions are calculated as:

$$S_i^+ = \sum_{j=1}^n w_j s_k(\alpha_{ij}, \alpha_j^+), \quad S_i^- = \sum_{j=1}^n w_j s_k(\alpha_{ij}, \alpha_j^-) \quad (21)$$

where, s_k denotes the selected sine-based similarity kernel.

The ranking index of alternative A_i is then defined as:

$$R_i = \frac{S_i^+}{S_i^+ + S_i^-}, \quad 0 \leq R_i \leq 1 \quad (22)$$

A larger R_i indicates that the alternative is more similar to the positive ideal profile relative to the negative ideal profile. The alternatives are therefore ranked in descending order of R_i .

For the energy-planning applications considered in this study, the ideal profile may simultaneously represent desirable characteristics such as higher energy recovery, lower carbon intensity, stronger grid flexibility, greater technology readiness, higher social acceptance, and lower cost.

4.5 Layer 5: Decision Separation and Robustness Indicators

A ranking alone does not indicate whether the leading alternative is clearly separated from its closest competitor. The fifth layer therefore evaluates both local and aggregate separation among the ranked alternatives.

Let $A_{(1)}$ and $A_{(2)}$ denote the first- and second-ranked alternatives, with ranking indices $R_{(1)}$ and $R_{(2)}$, respectively. The adjacent gap is defined as:

$$G = R_{(1)} - R_{(2)} \quad (23)$$

A larger value of G indicates stronger separation between the two leading alternatives, whereas a small value indicates that the decision is close and may be sensitive to relatively small changes in the underlying evaluations.

To complement the adjacent gap, the ASI measures the cumulative separation of the leading alternative from all remaining alternatives:

$$ASI = \sum_{i \neq a} (R_a - R_i) \quad (24)$$

where, a denotes the index of the highest-ranked alternative.

The ASI is an aggregate separation measure rather than a probability and is therefore not restricted to the interval $[0,1]$. A relatively large ASI indicates that the leading alternative is well separated from the decision set as a whole, but it does not necessarily imply strong separation from the second-ranked alternative. The ASI should therefore be interpreted together with the adjacent gap G .

For example, if $R_{(1)} = 0.613$ and $R_{(2)} = 0.605$, the adjacent gap is only $G = 0.008$, indicating a nearly tied decision even if the ASI is moderately large because of greater separation from lower-ranked alternatives.

The two separation measures therefore provide complementary information: G reflects the distinction between the two leading alternatives, while the ASI reflects the overall separation of the leader from the remaining decision set. Rank robustness is then evaluated separately through Monte Carlo perturbation in Section 4.6.

4.6 Monte Carlo Rank-Stability Analysis

The robustness of the rankings is further evaluated using Monte Carlo perturbation.

Let the original fuzzy evaluation of alternative i under criterion j be represented by A_{ij} .

A perturbed evaluation can be written generically as:

$$A_{ij}^{(b)} = A_{ij} + \varepsilon_{ij}^{(b)} \quad (25)$$

where, b denotes the simulation run and $\varepsilon_{ij}^{(b)}$ is a random perturbation bounded by the specified noise level.

After enforcing the corresponding q -rung or T -spherical admissibility condition, the ranking procedure is repeated for each simulation.

For B simulation runs, the probability that the original leading alternative remains first is estimated as:

$$P_{\text{leader}} = \frac{1}{B} \sum_{b=1}^B I \left[\text{rank}^{(b)}(A_{(1)}) = 1 \right] \quad (26)$$

where, $I[\cdot]$ is the indicator function.

Overall rank consistency can also be assessed through the Spearman rank correlation coefficient,

$$\rho_s = 1 - \frac{6 \sum_{i=1}^m d_i^2}{m(m^2 - 1)} \quad (27)$$

where, d_i is the difference between the original and perturbed ranks of alternative i .

These measures distinguish between three situations:

- a clearly preferred and robust alternative;
- a nominal leader whose rank is sensitive to uncertainty; and
- a genuinely contested decision for which the available information does not justify a unique recommendation.

This distinction is especially important for sustainable energy infrastructure, where a ranking decision may influence investments and operating commitments over many years.

4.7 Integration With the Energy Management System

As shown in Figure 2, the proposed framework can be connected conceptually with an EMS as a supervisory decision-support layer. Operational measurements and updated planning information enter the framework through the data-acquisition layer, after which they are normalized, represented within the appropriate fuzzy-information domain, weighted, and compared with the relevant ideal or reference profiles.

The resulting output is not limited to the identity of the highest-ranked alternative. The operator console or EMS receives the ranking together with the adjacent separation between the leading alternatives, the ASI, and rank-stability information where such analysis is available. This combination allows a nominally first-ranked alternative to be distinguished from a well-separated and robust recommendation.

The framework is intended to operate iteratively. Operational and planning information first enters the data-acquisition layer, after which it is fuzzified, weighted, and evaluated through the similarity-based reasoning process. The resulting rankings and robustness indicators are then returned to the operator console or EMS as decision-support information.

When updated information becomes available, such as changes in waste composition, electricity prices, heat demand, grid conditions, forecast uncertainty, technology performance, or stakeholder assessments, the decision matrix can be revised and the evaluation repeated. In this way, each new assessment cycle is informed by the latest operating and planning conditions rather than relying on a fixed decision state.

The appropriate update frequency depends on the type of decision. Strategic choices such as conversion technology, facility location, or industrial-symbiosis partnership would normally be reconsidered only when substantial new information becomes available. Operational choices, such as the preferred CHP operating strategy, can be reassessed more frequently as market, demand, or grid conditions evolve.

Machine-learning models may be incorporated upstream when additional feature extraction or forecasting is required. For example, operational traces or other high-dimensional data may first be transformed into compact representations before entering the fuzzification layer, while forecast-error information can be propagated into subsequent robustness analysis. The machine-learning component remains optional and does not replace the interpretable similarity-based decision layer.

Accordingly, the EMS connection shown in Figure 2 should be interpreted as a framework for iterative supervisory decision support rather than as a claim of fully autonomous real-time plant control.

5 Case Studies

The proposed framework is evaluated through four illustrative case studies representing different stages of sustainable WtE planning and operation. The cases were constructed to reflect the type of information that may be encountered in a mid-sized urban energy system, including expert assessments, waste-stream characteristics, technology indicators, siting conditions, and operational priorities.

The case studies are intended to evaluate the behavior, interpretability, and robustness of the proposed decision framework rather than to reproduce the conditions of a specific operating plant. Accordingly, the numerical inputs should be interpreted as structured test scenarios for methodological validation.

The four cases cover five interconnected decisions: waste-valorization pathway selection, district-to-pathway matching, plant siting, conversion-route selection, operating-strategy selection for a biogas CHP unit, and industrial-symbiosis partner selection.

5.1 Case 1: Waste-Valorization Pathway Selection, District Matching, and Plant Siting

5.1.1 Energy-planning context

The first case considers the early planning stage of an urban WtE system. A mid-sized coastal city must determine how MSW should be valorized, how different waste-producing districts should be matched with suitable treatment or energy-recovery pathways, and where the preferred facility should be located.

The six candidate pathways are:

- anaerobic digestion (AD);
- composting (CO);
- pyrolysis (PY);
- WtE incineration;
- material recovery facility (MRF); and
- mechanical-biological treatment (MBT).

These alternatives represent different positions within an integrated waste and energy-management system. AD, PY, and WtE involve direct energy recovery, whereas CO, material recovery, and MBT emphasize material stabilization, recycling, or preprocessing and may influence the quantity and composition of the residual stream subsequently available for energy conversion.

They are therefore evaluated here as alternative municipal waste-valorization strategies rather than as strictly equivalent energy-conversion technologies.

This distinction is important because the objective of the case is not simply to identify the conversion technology with the highest energy output. Instead, the city must balance energy recovery with capital and operating costs, greenhouse-gas reduction, material recovery, land requirements, public acceptance, and regional technology maturity.

The eight evaluation criteria are:

$$\begin{aligned}
 C_1 &= \text{capital cost} \\
 C_2 &= \text{operating cost} \\
 C_3 &= \text{net greenhouse-gas reduction} \\
 C_4 &= \text{energy recovery} \\
 C_5 &= \text{material recovery} \\
 C_6 &= \text{land requirement} \\
 C_7 &= \text{social acceptance}
 \end{aligned} \tag{28}$$

and

$$C_8 = \text{regional maturity} \tag{29}$$

Among these criteria, C_1 , C_2 , and C_6 are treated as cost criteria, whereas the remaining criteria are benefit criteria.

5.1.2 Fuzzy representation of the pathway decision

The complete decision matrix is given in Table 2. The evaluations are expressed as membership and non-membership pairs.

For all 48 entries in the decision matrix, the intuitionistic condition $\mu + \nu \leq 1$ is violated. However, the evaluations satisfy the Pythagorean condition $\mu^2 + \nu^2 \leq 1$, with the largest squared sum remaining within the admissible domain.

Accordingly, the decision information is represented using Pythagorean fuzzy numbers, corresponding to the q -rung orthopair formulation with $q = 2$. This representation allows strong but conflicting assessments to be retained rather than artificially rescaling the original expert information.

Table 2. Pythagorean fuzzy decision matrix for the municipal waste-valorization pathways

Pathway	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
AD	(0.55, 0.60)	(0.50, 0.55)	(0.80, 0.35)	(0.70, 0.45)	(0.60, 0.50)	(0.60, 0.55)	(0.75, 0.40)	(0.85, 0.30)
CO	(0.30, 0.80)	(0.35, 0.75)	(0.65, 0.50)	(0.20, 0.85)	(0.70, 0.45)	(0.75, 0.45)	(0.85, 0.30)	(0.90, 0.25)
PY	(0.75, 0.45)	(0.70, 0.50)	(0.60, 0.55)	(0.80, 0.40)	(0.55, 0.60)	(0.45, 0.70)	(0.50, 0.65)	(0.55, 0.60)
WtE	(0.90, 0.30)	(0.80, 0.40)	(0.45, 0.70)	(0.90, 0.30)	(0.30, 0.80)	(0.55, 0.60)	(0.35, 0.80)	(0.90, 0.25)
MRF	(0.45, 0.70)	(0.55, 0.60)	(0.60, 0.55)	(0.15, 0.90)	(0.90, 0.30)	(0.40, 0.75)	(0.80, 0.35)	(0.85, 0.30)
MBT	(0.65, 0.55)	(0.65, 0.55)	(0.70, 0.45)	(0.55, 0.60)	(0.75, 0.45)	(0.65, 0.50)	(0.65, 0.50)	(0.75, 0.40)

Note: C_1 = capital cost; C_2 = operating cost; C_3 = net greenhouse-gas reduction; C_4 = energy recovery; C_5 = material recovery; C_6 = land requirement; C_7 = social acceptance; C_8 = regional maturity; AD = anaerobic digestion; CO = composting; PY = pyrolysis; WtE = waste-to-energy; MRF = material recovery facility; MBT = mechanical-biological treatment. C_1 , C_2 , and C_6 are cost criteria, for which μ represents the degree to which the corresponding cost or requirement is judged high.

5.1.3 Entropy weights and pathway ranking

Entropy weighting assigns the largest weights to regional maturity and energy recovery, with $w_{C_8} = 0.179$ and $w_{C_4} = 0.168$, respectively. As shown in Figure 3a, these two criteria provide the strongest discrimination among the six valorization pathways, whereas operating cost, greenhouse-gas reduction, and land requirement receive comparatively smaller weights.

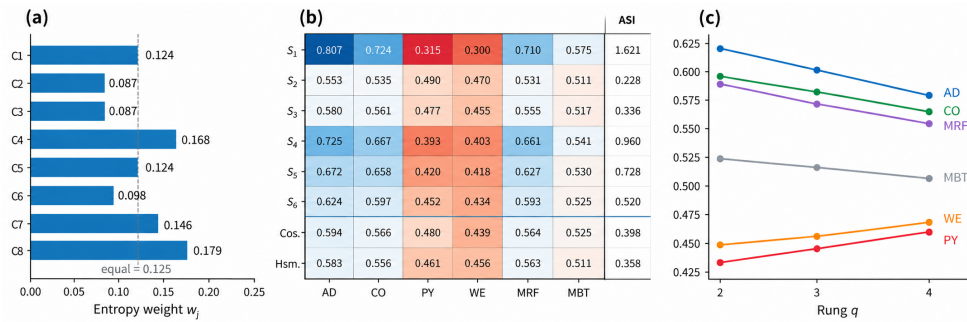


Figure 3. Pathway-selection results: (a) entropy-derived criterion weights; (b) pathway rankings across the six similarity measures and two benchmark measures; (c) ranking sensitivity to the rung parameter

Note: ASI = Aggregate Separation Index; AD = anaerobic digestion; PY = pyrolysis; CO = composting; WtE = waste-to-energy; MRF = material recovery facility; MBT = mechanical-biological treatment; Cos. = cosine similarity; Ham. = Hamming similarity.

The relatively high weight assigned to C_4 shows that energy recovery is one of the strongest discriminatory criteria in the pathway decision, while the high value of C_8 indicates that local technological and institutional maturity also contributes substantially to the differentiation of alternatives.

Land requirement, greenhouse-gas reduction, and operating cost receive smaller entropy weights because the alternatives are less strongly separated on these criteria.

The six proposed similarity measures and the two benchmark measures produce the same first four positions, with AD ranked first, followed by CO, MRF, and MBT. Figure 3b shows that the leading alternative remains unchanged across all eight measures, although the relative positions of the two lower-ranked thermal pathways vary with the selected kernel. The accompanying ASI values further indicate that the degree of separation between the leading alternative and the remaining pathways differs substantially across the similarity measures.

Under the reference kernel used for the principal analysis, the ranking indices of the three leading alternatives are $R_{AD} = 0.624$, $R_{CO} = 0.597$, and $R_{MRF} = 0.593$.

Thus, the ranking gaps are $R_{AD} - R_{CO} = 0.027$, and $R_{AD} - R_{MRF} = 0.031$.

The ranking therefore identifies AD as the preferred pathway, but the difference between CO and material recovery remains relatively small.

The result should not be interpreted as evidence that AD is universally superior to thermal WtE technologies. Instead, it reflects the joint influence of the eight criteria and the specific evaluation matrix used in this scenario. From an energy-sustainability perspective, AD performs well because it combines recoverable energy from biogas with relatively favorable assessments on cost, greenhouse-gas reduction, land requirement, and social acceptance. Thermal

WtE performs strongly on energy recovery and regional maturity but is penalized by less favorable evaluations on capital cost, material recovery, and social acceptance.

This result illustrates why a sustainable energy decision cannot be based on energy yield alone.

5.1.4 Sensitivity to the rung and weighting scheme

The first four positions remain unchanged over the tested range of the rung parameter. As illustrated in Figure 3c, AD remains the leading pathway as q increases from 2 to 4, while CO, MRF, and MBT also retain their relative positions. The only change occurs between PY and WtE at the lower end of the ranking, indicating that the leading decision is substantially less sensitive to the rung parameter than the trailing alternatives.

Changes in the weighting scheme have a larger effect.

Under equal criterion weights, the ranking of the three leading alternatives becomes $R_{AD} = 0.610$, $R_{MRF} = 0.604$, and $R_{CO} = 0.593$. Thus, $AD > MRF > CO$.

This result shows that the relative position of CO and material recovery depends more strongly on criterion priorities than on the choice of similarity kernel.

The difference can be traced primarily to the entropy weighting of energy recovery. Because MRF performs relatively weakly on C_4 , its ranking improves when the additional influence assigned to energy recovery by the entropy scheme is removed.

This sensitivity has a direct planning interpretation. If the municipality gives greater priority to recoverable energy and the integration of waste resources into the local energy system, CO is preferred to material recovery in the present scenario. If all sustainability dimensions are treated equally, material recovery becomes more competitive.

The method therefore does not hide the policy trade-off behind a single numerical ranking; instead, it identifies the criterion priorities responsible for the change.

5.1.5 District-to-pathway matching

The second part of Case 1 examines whether different urban districts should be associated with different wastevalorization profiles rather than assuming that a single treatment route is suitable for the entire city.

Four districts are characterized using eight indicators describing waste composition, moisture content, generation intensity, collection coverage, and urban density.

The district profiles are compared with three reference pathway profiles representing:

- biological treatment;
- thermal conversion; and
- material recovery.

After fuzzification, all six similarity kernels produce the same district assignments.

The old urban core and the peri-urban fringe contain organic fractions exceeding 60% and are therefore matched with the biological pathway.

The industrial district has a plastic fraction of approximately 30% and is matched with the thermal pathway.

The coastal residential district is assigned to the material-recovery profile.

As shown in Figure 4a, the old urban core and peri-urban district are most similar to the biological-treatment prototype, while the industrial district is matched with the thermal pathway and the coastal residential district with the material-recovery profile. The coastal district shows the weakest separation among the four assignments, with an ASI of 0.660, indicating comparatively greater ambiguity in its pathway match.

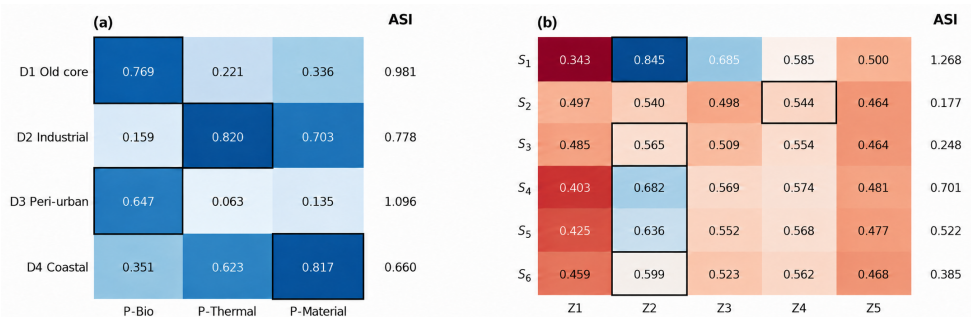


Figure 4. District-to-pathway matching and AD plant siting results: (a) district-to-pathway matching under S_6 ; (b) AD plant siting across the six similarity measures

Note: ASI = Aggregate Separation Index; AD = anaerobic digestion.

The latter assignment also has the smallest ASI among the four districts. Its similarity to the selected material-recovery profile is 0.817, whereas its similarity to the thermal profile is 0.623. The corresponding ASI is 0.660.

Across the tested fuzzification range, no district changes its assigned pathway, and each ASI value changes by less than 0.05.

Across the tested fuzzification range, no district changes its assigned pathway, and each confidence value changes by less than 0.05.

This result demonstrates that the framework can support spatially differentiated waste-energy planning. Rather than treating the municipal waste stream as homogeneous, the system can use district-level characteristics to identify which areas are more suitable for biological treatment, thermal energy recovery, or material recovery.

Such differentiation can improve the consistency between feedstock quality and downstream conversion technology.

5.1.6 Plant siting

The final part of Case 1 considers the location of an AD facility.

Five candidate zones are evaluated using six siting criteria. The criteria reflect both conventional infrastructure requirements and energy-system integration conditions.

Zone Z_2 combines favorable grid access and heat-offtake conditions with relatively low land cost and low environmental sensitivity.

It therefore ranks first under five of the six similarity kernels.

Figure 4b shows that Z_2 is the leading site under five of the six similarity measures. Only S_2 ranks Z_4 slightly higher, with a difference of 0.004 between the two sites. The result therefore supports Z_2 as the most consistently preferred location, while also showing that the site decision is less strongly separated than the pathway-selection result.

One flatter kernel places Z_4 slightly ahead of Z_2 , with a difference of only 0.004.

The second-ranked site varies between Z_3 and Z_4 .

The result indicates that Z_2 is the most consistently preferred site, although the alternative site ranking is less stable than the pathway ranking.

From an energy-system perspective, this is an important result because WtE siting should not be evaluated solely in terms of land availability or environmental constraints. Proximity to grid infrastructure and usable heat demand can directly affect the fraction of recovered energy that can be productively utilized.

A site with favorable heat-offtake conditions may support CHP operation more effectively than a site that is cheaper but poorly connected to thermal demand.

Accordingly, the siting component of the framework links spatial planning with the practical utilization of recovered electricity and heat.

5.1.7 Interpretation of Case 1

Case 1 demonstrates three features of the proposed framework.

First, the preferred waste-valorization strategy depends on a combination of energy, environmental, economic, technical, and social criteria rather than on energy recovery alone.

Second, district-level differences in waste composition can be incorporated into the planning process, allowing different parts of the city to be associated with different valorization pathways.

Third, the facility-siting decision incorporates grid and heat-offtake considerations, thereby connecting waste-management planning directly with energy-system integration.

Taken together, these results show that sustainable WtE planning is not a single technology-selection problem. It involves a linked sequence of decisions concerning feedstock characteristics, conversion pathway, spatial allocation, infrastructure access, and the utilization of recovered energy.

5.2 Case 2: Energy-Conversion Route Selection Under Stakeholder Abstention

5.2.1 Energy-planning context

The second case considers the selection of an energy-conversion route for the municipal waste and biogenic resource stream. Unlike the first case, which compares broader waste-valorization strategies, this case focuses specifically on technologies and energy pathways capable of producing useful heat, electricity, gaseous fuel, or hydrogen.

Five candidate conversion routes are evaluated:

1. biogas CHP with district-heat utilization;
2. biomethane injection into the gas network;
3. refuse-derived fuel (RDF) co-firing in a cement kiln;
4. WtE incineration with CHP; and
5. syngas-to-hydrogen conversion.

The alternatives represent different forms of energy recovery and therefore differ not only in cost and technological maturity, but also in the type of energy carrier produced, their ability to support the local energy system, and their potential contribution to carbon reduction.

Seven criteria are considered: levelized cost, grid flexibility, carbon intensity, energy-offtake security, public acceptance, technology readiness, and revenue-stacking potential.

These criteria capture both conventional energy-system performance and implementation feasibility. In particular, grid flexibility reflects the ability of a conversion route to contribute to balancing or dispatchable operation, while offtake security represents the likelihood that the recovered electricity, heat, gas, or fuel can be continuously utilized. Revenue stacking accounts for the possibility of obtaining value from more than one energy or ancillary-service stream.

5.2.2 Representation of stakeholder abstention

The evaluations are expressed as support-abstention-opposition triplets.

In the present case, the largest linear sum of the three information components reaches 1.45. Consequently, the evaluations cannot be represented within the conventional picture-fuzzy domain.

The largest squared sum is 0.895, which satisfies the admissibility condition of the selected T -spherical representation. A T -spherical fuzzy model is therefore used so that stakeholder abstention can be retained explicitly rather than being redistributed between support and opposition.

This distinction is relevant to energy-infrastructure planning. Stakeholders may support a technology because of its carbon or energy benefits while remaining uncertain about its commercial maturity, local environmental implications, or infrastructure requirements. Treating such neutrality as either positive or negative evidence would artificially increase the apparent certainty of the ranking.

5.2.3 Criterion weights

Entropy weighting assigns the highest importance to technology readiness, with a weight of 0.220, followed by revenue-stacking potential, with a weight of 0.163.

This result indicates that the largest differences among the candidate routes arise from their maturity and their ability to generate value through multiple energy or service streams.

The importance of technology readiness is particularly relevant in municipal WtE planning because a technically attractive conversion route may still present substantial implementation risk if large-scale deployment experience is limited.

Revenue stacking is also important because WtE assets increasingly operate within integrated energy systems rather than as isolated waste-treatment facilities. A conversion route capable of supplying electricity, recoverable heat, renewable gas, fuel, or flexibility services may have a more resilient economic position than one depending on a single revenue stream.

5.2.4 Conversion-route ranking

All six similarity measures identify biogas CHP as the leading energy-conversion route.

Biomethane injection ranks second, followed by WtE-CHP.

The leading position of biogas CHP is therefore not dependent on the choice of similarity kernel. Under the reference measure, the ranking index for biogas CHP is 0.662, while the corresponding ASI is 0.539.

The ranking result reflects the balance between energy-system performance and implementation conditions contained in the decision matrix. Biogas CHP can simultaneously provide dispatchable electricity and useful thermal energy and can therefore contribute to both local energy recovery and operational flexibility when a stable heat sink is available.

Biomethane injection provides a different form of system integration by converting biogas into a storable and transportable energy carrier that can be introduced into the gas network. Its ranking indicates that it remains a competitive route, although its performance under the present criterion structure does not exceed that of CHP. WtE-CHP provides substantial direct energy recovery and can support both electricity and heat supply, but its overall ranking is affected by the combined evaluation of cost, carbon intensity, public acceptance, and technology-related criteria.

The result should therefore not be interpreted as a universal technological hierarchy. It identifies the preferred route only for the present decision matrix and criterion-weight structure.

5.2.5 Effect of stakeholder abstention

To examine whether the explicit abstention component materially affects the decision, the neutral component is removed and each triplet is reduced to a Pythagorean pair.

The ranking order remains unchanged: biogas CHP remains first, biomethane remains second, and WtE-CHP remains third.

As shown in Figure 5a, removing the abstention component does not alter the ordering of the leading conversion routes, although the ranking indices shift slightly. This indicates that stakeholder abstention affects the separation among alternatives more strongly than the identity of the highest-ranked route in the present case.

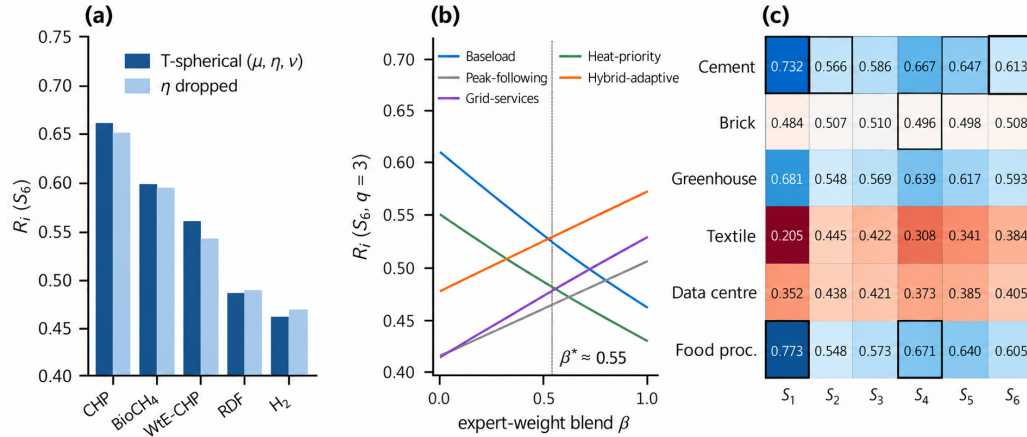


Figure 5. Conversion-route, CHP operating-strategy, and industrial-symbiosis partner results: (a) conversion-route ranking with and without stakeholder abstention; (b) CHP operating-strategy ranking under different expert-weight blends; (c) industrial-symbiosis partner rankings across the six similarity measures

Note: BioCH₄ = biological methane; WtE = waste-to-energy; RDF = refuse derived fuel; H₂ = Hydrogen; CHP = combined heat and power.

However, the ranking indices change by as much as 0.020, while the ASI values change by as much as 0.026.

This result is important because it shows that stakeholder abstention does not necessarily determine which energy technology is ranked first, but it can alter the apparent separation between competing technologies. In practical terms, ignoring abstention may therefore create excessive confidence in a decision even when the final ranking itself appears unchanged.

For long-lived energy infrastructure, this distinction is significant. A municipality may reach the same technology preference under two uncertainty representations, while the strength of evidence supporting that preference can be materially different.

5.2.6 Implications for sustainable energy planning

Case 2 demonstrates that the proposed framework can evaluate energy-conversion pathways that provide different forms of useful energy while retaining uncertainty in stakeholder assessments.

The result also illustrates the importance of evaluating WtE technologies as components of an integrated energy system rather than solely as waste-disposal technologies.

The criteria considered in this case connect waste conversion with several dimensions central to energy sustainability: carbon performance, grid flexibility, technology maturity, energy-offtake reliability, and the ability to obtain value from multiple energy streams.

The preference for biogas CHP in the present scenario is therefore driven by its combined performance across these dimensions rather than by a single criterion.

More importantly, the comparison with and without the abstention component shows that uncertainty affects not only the ordering of technologies but also the degree of separation among the leading alternatives. This provides decision makers with additional information when deciding whether the available evidence is sufficiently strong to justify investment in a particular conversion route.

5.3 Case 3: Operating Strategy for the Biogas Combined Heat and Power Unit

5.3.1 Operational energy-management context

The third case considers short- to medium-term operation of a biogas CHP unit after the conversion technology has been selected.

Unlike the previous cases, which focus mainly on strategic planning, this case addresses an operational decision: how the CHP unit should be dispatched when economic return, grid support, emissions, equipment wear, forecast uncertainty, and operational complexity cannot all be optimized simultaneously.

Five candidate operating strategies are considered:

- Baseload;
- Peak-following;
- Grid-services;

- Heat-priority; and
- Hybrid-adaptive operation.

These strategies represent different operating philosophies within an integrated energy system.

Baseload operation prioritizes stable and continuous energy production. Peak-following operation responds more strongly to electricity-price or demand peaks. Grid-services operation gives greater emphasis to flexibility and system-support functions. Heat-priority operation follows thermal demand more closely, while the Hybrid-adaptive strategy attempts to balance energy-market, grid, and operational conditions dynamically.

The decision therefore reflects a central challenge in sustainable CHP operation: the most profitable strategy is not necessarily the strategy with the lowest operational risk, the lowest emissions, or the highest thermal-utilization efficiency.

5.3.2 Decision criteria and uncertainty representation

The five operating strategies are evaluated using six criteria:

$$\begin{aligned}
C_1 &= \text{revenue} \\
C_2 &= \text{grid-stability contribution} \\
C_3 &= \text{emission intensity} \\
C_4 &= \text{equipment wear} \\
C_5 &= \text{sensitivity to forecast error}
\end{aligned} \tag{30}$$

and

$$C_6 = \text{operational complexity} \tag{31}$$

Revenue and grid-stability contribution are benefit criteria, while emission intensity, equipment wear, forecast sensitivity, and operational complexity are treated as cost criteria.

The evaluations cannot be represented adequately within the Pythagorean domain. The largest squared membership–non-membership sum exceeds unity, $\mu^2 + \nu^2 > 1$, whereas the corresponding Fermatean condition remains admissible: $\mu^3 + \nu^3 \leq 1$.

The decision matrix is therefore represented using Fermatean fuzzy numbers, corresponding to $q = 3$.

This higher-rung representation allows relatively strong support and opposition to coexist without rescaling the original evaluations.

In practical terms, this is appropriate for CHP operation because an operating strategy can simultaneously have strong advantages and disadvantages. A grid-responsive strategy, for example, may offer high revenue and valuable flexibility while also increasing wear, forecast exposure, or operational complexity.

5.3.3 Ranking under entropy-derived weights

Under entropy-derived criterion weights, the risk-related criteria receive relatively strong influence because they provide substantial discrimination among the five operating strategies.

The Baseload strategy ranks first under all six similarity kernels.

The principal ranking result is therefore robust with respect to the choice of similarity measure.

Under the reference kernel, the Baseload strategy obtains a ranking index of $R_{\text{Baseload}} = 0.608$, with an ASI of 0.578.

Heat-priority is reported as the second-ranked alternative in the summary results, while Peak-following and Grid-services remain very close under several parameter settings.

The leading position of Baseload under entropy weighting reflects the balance between energy production and operational risk contained in the decision matrix.

Although more responsive strategies may provide greater opportunities for market revenue or grid support, they are also exposed more strongly to forecast error, equipment cycling, and operational complexity. Because the entropy weights give substantial influence to these differentiating risk criteria, the more conservative Baseload strategy becomes preferable.

This result should therefore not be interpreted as evidence that continuous Baseload operation is inherently the most sustainable CHP strategy. It is the preferred strategy only under the present data and the objective information structure generated by the entropy-weighting procedure.

5.3.4 Effect of expert energy-system priorities

Operational decisions are often influenced by objectives that cannot be derived solely from the dispersion of the observed data.

For this reason, the entropy-derived weights are blended with an expert weight vector that assigns greater importance to revenue and grid-stability contribution.

In the expert weight vector, $w_{\text{revenue}} = 0.30$ and $w_{\text{grid}} = 0.30$.

Let the blended criterion-weight vector be $\mathbf{w}(\lambda) = (1 - \lambda) \mathbf{w}^{(E)} + \lambda \mathbf{w}^{(S)}$, where $0 \leq \lambda \leq 1$.

Here, $\lambda = 0$ corresponds entirely to entropy-derived weights, while larger values of λ progressively increase the influence of the expert priorities.

At the lower expert-weight contribution tested in the original analysis, Baseload remains first, with a ranking index of approximately 0.535, compared with 0.525 for Hybrid-adaptive operation. The corresponding confidence falls substantially, while the ASI decreases to 0.191, indicating that the decision becomes much less distinct once greater emphasis is placed on revenue and grid contribution.

As the expert-weight contribution increases further, the two strategies cross.

Figure 5b illustrates how the preferred CHP operating strategy changes as the expert-weight blend increases. Baseload is favored when entropy-derived weights dominate, whereas Hybrid-adaptive becomes preferable once greater importance is assigned to revenue and grid-stability contribution.

At the higher expert-weight setting reported in the original results, Hybrid-adaptive becomes the leading strategy with a ranking index of 0.571 while Baseload falls to fourth position. This ranking reversal is one of the most important results of the case.

It shows that the preferred CHP operating strategy depends not only on uncertain operational data, but also on the relative priority assigned to conservative operation versus active participation in energy and grid-service markets.

5.3.5 Interpretation of the operational trade-off

The transition from Baseload to Hybrid-adaptive operation reflects a change in the priorities underlying the energy-management decision. Under entropy-derived weights, the ranking favors operational stability and lower exposure to forecast uncertainty, equipment wear, and operating complexity. As greater importance is assigned to revenue and grid-stability contribution, the more flexible Hybrid-adaptive strategy becomes increasingly competitive and eventually overtakes Baseload.

The two strategies therefore represent different operating priorities. Baseload operation provides greater predictability and limits the need for frequent adjustments in response to short-term market or system fluctuations. By contrast, Hybrid-adaptive operation offers greater flexibility to respond to electricity prices, grid conditions, and changes in energy demand, but this flexibility is accompanied by higher operational complexity and potentially greater exposure to uncertain operating conditions.

This distinction is particularly relevant to biogas CHP because the unit can provide several energy-system services simultaneously, including electricity generation, useful heat, dispatchable output, and grid-support capability. The preferred operating strategy therefore depends on whether the plant is expected primarily to deliver stable energy production or to participate more actively in flexible and market-responsive operation.

5.3.6 Implications for smart-energy operation

Case 3 demonstrates that the proposed framework can support operational decisions in addition to long-term infrastructure planning.

The result also highlights why a single static criterion-weight vector may be insufficient for smart-energy operation.

During periods in which equipment reliability, forecast uncertainty, or stable heat supply are the dominant concerns, a conservative operating strategy may be appropriate.

During periods in which grid flexibility and market value become more important, a more adaptive strategy may be preferable.

This creates a natural connection with the energy-management-system loop introduced in Section 4. As electricity prices, thermal demand, grid conditions, or forecast uncertainty change, the input information and criterion priorities can be updated and the operating strategy can be re-evaluated.

Accordingly, the proposed decision framework can be interpreted as a supervisory decision layer above the conventional CHP control system rather than as a replacement for real-time process control.

The role of the framework is to determine which operating philosophy is most appropriate under the current technical and strategic conditions, while the lower-level control system continues to manage equipment-level variables and constraints.

5.3.7 Main finding of Case 3

The principal result of Case 3 is not simply that Baseload obtains the highest score under one weighting scheme. More importantly, the analysis identifies the conditions under which the preferred operating strategy changes.

Under the objective information structure generated by entropy weighting, Baseload is the most stable choice. When revenue and grid-support objectives receive substantially greater importance, Hybrid-adaptive becomes preferable.

The case therefore demonstrates that uncertainty-aware decision support can expose an energy-management trade-off that would be hidden by reporting only a single final ranking.

5.4 Case 4: Industrial-Symbiosis Partner Selection

5.4.1 Energy and resource-integration context

The fourth case considers the selection of an industrial-symbiosis partner capable of utilizing outputs from the WtE system.

The objective is not limited to identifying a nearby industrial facility. Instead, the decision concerns whether recovered heat, digestate, RDF, and related by-products can be integrated into surrounding industrial processes in a manner that improves overall resource and energy utilization.

Six candidate facilities are evaluated using six criteria:

$$\begin{aligned} C_1 &= \text{resource match} \\ C_2 &= \text{by-product uptake} \\ C_3 &= \text{distance} \\ C_4 &= \text{contract stability} \\ C_5 &= \text{emission-credit potential} \end{aligned} \tag{32}$$

and

$$C_6 = \text{integration cost} \tag{33}$$

No candidate performs best on all six criteria.

The resulting decision therefore represents a typical industrial-symbiosis problem in which technical compatibility, logistics, commercial reliability, carbon implications, and integration costs must be considered simultaneously.

The decision matrix satisfies the Pythagorean admissibility condition and is therefore represented using $q = 2$.

5.4.2 Criterion weighting and ranking

Entropy weighting assigns the largest weight to by-product uptake, $w_{C_2} = 0.251$.

This result indicates that the ability of a potential partner to absorb recovered material or energy streams provides the strongest discrimination among the six candidates.

The cement plant performs particularly well on this criterion and ranks first under four of the six similarity kernels.

The food processor ranks first under the remaining two kernels.

As shown in Figure 5c, the leading industrial-symbiosis partner depends on the selected similarity measure. The cement plant ranks first under four kernels, while the food processor leads under the remaining two, confirming that the partner-selection problem is substantially less decisive than the pathway and conversion-route decisions.

The greenhouse cluster remains close to the two leading alternatives but does not consistently overtake them.

For the recommended kernels, the maximum difference between the two leading alternatives is only 0.008. Under the reference measure, the reported scores are approximately $R_{\text{cement}} = 0.613$, and $R_{\text{food}} = 0.605$. Thus, $R_{\text{cement}} - R_{\text{food}} = 0.008$.

This difference is too small to support a strong practical distinction between the two alternatives.

5.4.3 Energy interpretation of the partner trade-off

The two leading candidates perform well for different reasons. The cement plant benefits from stronger by-product uptake potential, which is particularly relevant when RDF or other energy-bearing residuals can substitute part of the conventional fuel requirement of an industrial process. The food processor, by contrast, performs more favorably in terms of distance and integration cost.

The comparison therefore reflects a trade-off between utilization potential and integration burden. From a sustainable energy perspective, the partner-selection problem is not simply a matter of identifying the facility capable of accepting the largest quantity of residual material. A suitable partner should also provide a practical and economically viable pathway for recovered heat, fuel, or by-products to be utilized with limited transport requirements, infrastructure modifications, and contractual risk.

The preferred partner consequently depends on the relative importance assigned to resource uptake, proximity, and integration requirements. In the present case, this balance is sufficiently close that neither of the two leading candidates can be regarded as decisively superior on the basis of the available information.

5.4.4 Why the framework reports a contested decision

The reference analysis gives an ASI of 0.570. Taken alone, this aggregate value appears to indicate moderate overall separation from the remaining candidates. However, the adjacent gap between the two leading alternatives is only 0.008, and the identity of the leader changes across similarity kernels. The relatively large ASI is therefore driven partly by the weaker lower-ranked alternatives and should be interpreted together with the adjacent gap.

These two observations indicate that the available information does not support a robust separation of the cement plant and the food processor.

The framework therefore classifies the decision as contested and does not issue a unique partner recommendation. This is an important feature of the proposed method.

A decision-support system should not necessarily return a winner in every case. When two alternatives are effectively indistinguishable within the uncertainty of the available evaluations, reporting that ambiguity is more informative than presenting a nominal first-place score as a definitive result.

In the present case, improved information on actual offtake capacity would be particularly valuable.

For example, a more reliable estimate of long-term heat or by-product demand, supported by a contractual commitment or verified operating requirement, could materially alter the ranking.

5.4.5 Implications for industrial energy symbiosis

The industrial-symbiosis case extends the proposed framework beyond plant-level energy conversion.

Recovered energy and material streams provide sustainability value only when they can be utilized effectively downstream.

For example, the theoretical availability of excess heat does not guarantee an energy benefit if no suitable thermal user is located within an economically viable distance.

Similarly, RDF has limited practical value if the receiving process cannot absorb it consistently or if integration costs are excessive.

The partner-selection layer therefore connects WtE operation with broader industrial energy integration.

In this sense, the proposed framework evaluates not only how energy is recovered, but also whether the recovered energy and associated material streams can be embedded within a wider circular energy system.

5.5 Monte Carlo Rank-Stability Analysis

5.5.1 Purpose of the robustness analysis

A deterministic ranking provides only a single representation of the decision problem.

In practical WtE planning, however, the input evaluations are uncertain and may vary because of changes in expert judgment, waste characteristics, operating conditions, forecast error, or stakeholder information.

Monte Carlo perturbation is therefore used to assess whether the leading alternatives remain stable when the decision matrix is subjected to small changes.

The original analysis applies Monte Carlo testing to Cases 1, 3, and 4.

The perturbation magnitude is set to $\varepsilon = 0.05$, which corresponds to approximately half a step on the underlying ten-point evaluation scale used to construct the scenario data.

For each perturbed matrix, the complete ranking procedure is repeated.

The analysis focuses on three indicators:

1. probability that the original leader remains in first place;
2. probability that the complete ranking order remains unchanged; and
3. mean Spearman rank correlation between the original and perturbed rankings.

5.5.2 Stability of the leading alternatives

The pathway-selection case exhibits high leader stability.

Across the tested similarity kernels, the probability that the original leader remains first is at least $P_{\text{leader}} \geq 0.917$.

Under the reference measure reported in Table 3, $P_{\text{leader}} = 0.948$.

The CHP operating-strategy case is even more stable.

The leader remains first with probability of at least 0.983, and under the reference measure the reported probability reaches $P_{\text{leader}} = 0.999$.

As shown in Figure 6a, the pathway and CHP operating-strategy decisions exhibit high rank-1 stability under the $\varepsilon = 0.05$ perturbation, whereas the industrial-symbiosis partner decision is substantially less stable. This contrast is consistent with the much smaller separation between the two leading partner alternatives.

These results indicate that the preferred pathway and the preferred operating strategy under the corresponding weighting assumptions are not artifacts of very small changes in the input evaluations.

By contrast, the industrial-symbiosis partner decision is substantially less stable.

The probability that the nominal leader remains first falls to approximately 0.54-0.63, depending on the similarity kernel. Under the reference result reported in Table 3, $P_{\text{leader}} = 0.632$.

Table 3. Summary and robustness of the five decision problems under S_6 with entropy-derived criterion weights

Decision	Fuzzy Representation	Leader (R)	Runner-Up (R)	Adjacent Gap G	ASI	P_{leader}	Interpretation
Pathway	Pythagorean	AD (0.624)	CO (0.597)	0.027	0.520	0.948	Robust leader
Site	Pythagorean	Z2 (0.599)	Z4 (0.562)	0.037	0.385	–	Leader stable across most-kernels
Route	T -spherical	CHP (0.662)	BioCH ₄ (0.598)	0.064	0.539	–	Clear leader
Strategy	Fermatean	Baseload (0.608)	Heat-priority (0.551)	0.057	0.578	0.999	Depends on weight-blend β
Partner	Pythagorean	Cement (0.613)	Food processor (0.605)	0.008	0.570	0.632	Contested

Note: ASI denotes the Aggregate Separation Index and measures the cumulative separation of the leading alternative from all remaining alternatives. It is not a probability and is therefore not restricted to $[0,1]$. AD = anaerobic digestion; CHP = combined heat and power; BioCH₄ = biological methane; “–” = no data.

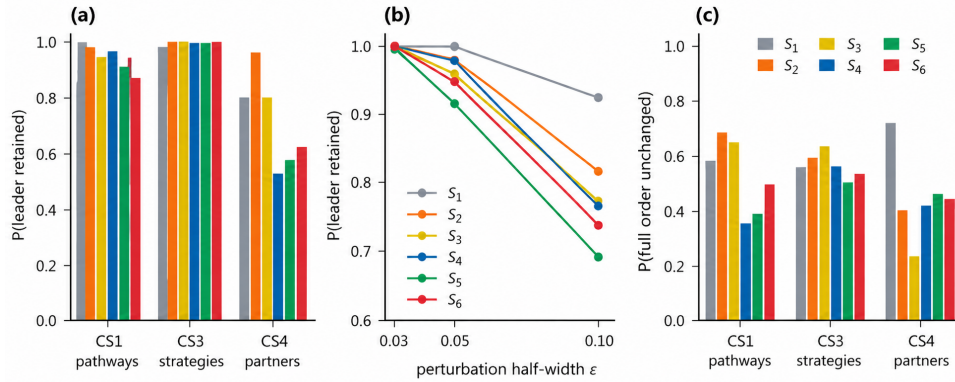


Figure 6. Monte Carlo rank-stability results: (a) rank-1 stability at $\epsilon = 0.05$; (b) pathway-selection stability under increasing perturbation; (c) full-order stability at $\epsilon = 0.05$

This result is consistent with the very small score difference between the cement plant and the food processor and confirms that the partner-selection problem should remain classified as contested.

5.5.3 Full-order stability

Although the leading alternative is stable in several cases, the complete ranking order is less robust. As illustrated in Figure 6c, full-order stability is considerably lower than rank-1 stability across the three cases. This indicates that lower-ranked alternatives frequently exchange nearby positions even when the leading alternative remains unchanged.

The probability that the full ranking remains unchanged ranges approximately from 0.24 to 0.72. This lower full-order stability results mainly from nearly tied alternatives in the middle or lower part of the ranking.

Accordingly, a stable first-place decision should not be interpreted as evidence that every position in the ranking is equally reliable.

This distinction is particularly important when the decision objective is to identify one preferred technology or strategy rather than to establish a definitive order for all alternatives.

5.5.4 Rank correlation under perturbation

Despite changes in some lower ranking positions, the mean Spearman rank correlation remains above 0.93. This indicates that the general ordering structure is preserved even when individual alternatives exchange nearby positions.

The result suggests that the proposed decision framework is relatively robust at the system level while still remaining sensitive enough to identify cases in which the top alternatives cannot be separated confidently.

5.5.5 Effect of noise intensity

The pathway case was also examined under different perturbation levels. Figure 6b shows that the rank-1 stability of the pathway decision decreases as the perturbation magnitude increases. The leading pathway remains highly stable at lower noise levels, but its survival probability declines progressively as the ratings are subjected to larger perturbations.

As the uncertainty magnitude increases, leader stability decreases.

At the lower perturbation level, the probability that the leader remains first is close to unity for several kernels.

When the perturbation magnitude increases to $\varepsilon = 0.10$, leader survival falls to approximately 0.69–0.93.

This result is expected because larger perturbations introduce more substantial changes into the underlying evaluation matrix.

More importantly, the analysis quantifies the extent to which a recommendation can tolerate uncertainty before its stability begins to deteriorate.

For long-term energy-infrastructure planning, such information is more useful than reporting only a deterministic ranking.

5.5.6 Influence of the similarity kernel

The Monte Carlo results also reveal differences among the similarity kernels.

In the pathway-selection case, the steeper kernels produce more frequent changes in the trailing ranks, whereas flatter kernels preserve the lower-order rankings more often.

The effect on the leading alternative is considerably smaller.

This result supports the earlier observation that the choice of criterion weights and the underlying decision data have a stronger influence on the main recommendation than the specific similarity kernel used within the admissible set.

Consequently, the framework should not be interpreted as producing materially different energy decisions merely because different valid similarity measures are available.

Instead, the kernel comparison acts primarily as an internal robustness test.

5.5.7 Summary of decision robustness

The Monte Carlo results distinguish clearly among the different types of decisions investigated in this study.

For the waste-valorization pathway, $P_{\text{leader}} = 0.948$, indicating a robust leading alternative.

For the CHP operating strategy, $P_{\text{leader}} = 0.999$, showing very strong stability under the specified perturbation level.

For the industrial-symbiosis partner, $P_{\text{leader}} = 0.632$, indicating that the nominal leader is not sufficiently stable to justify a definitive recommendation.

The robustness analysis therefore supports the central purpose of the framework: not only to identify the highest-ranked energy alternative, but also to determine whether the evidence supporting that ranking is sufficiently strong for practical decision making.

6 Machine-Learning Integration and Robustness Benchmark

6.1 Role of Machine Learning Within the Decision Framework

The proposed framework is primarily a decision-support system rather than a predictive machine-learning model. Its core function is to compare alternatives against interpretable energy-performance profiles while retaining uncertainty, hesitation, and separation information.

Machine learning can nevertheless be incorporated at several points without changing this decision logic.

First, data-driven models can be used upstream to extract compact features from operational data such as waste-stream images, SCADA signals, smart-meter records, or historical plant measurements.

Second, historical observations can be used to construct or update representative class prototypes.

Third, forecasting models can provide estimates of future variables such as waste availability, electricity prices, thermal demand, or operating conditions. Their forecast uncertainty can then be propagated into the decision layer through the perturbation and rank-stability analysis.

The role of machine learning is therefore complementary. It supplies data representations or forecasts, whereas the similarity-based fuzzy layer remains responsible for interpretable comparison, uncertainty handling, and decision-separation assessment.

This separation is important for sustainable energy applications because a highly accurate predictive model does not necessarily provide sufficient information to justify a long-term infrastructure or operating decision.

6.2 Benchmark Design

To examine the predictive cost of using the similarity-based decision layer, synthetic waste-stream profiles are generated from the three pathway prototypes used in the earlier district-matching analysis.

Gaussian noise is added at four levels to represent increasing uncertainty in the measured indicators.

For each class, the benchmark uses 60 training profiles and 300 test profiles. The experiment is repeated using 10 independent random seeds.

The similarity-based prototype matcher is compared with several conventional classification approaches:

- Euclidean prototype matching;
- cosine similarity;
- five-nearest neighbors;

- logistic regression;
- random forest [28];
- support vector machine; and
- linear discriminant analysis followed by fuzzy matching.

The conventional machine-learning benchmarks are implemented using the scikit-learn framework [29]. The same normalized input features are used across the competing methods to ensure that the comparison reflects the classification mechanism rather than differences in preprocessing.

6.3 Classification Performance Under Increasing Noise

Table 4 reports the mean test accuracy obtained across ten random seeds under four levels of indicator noise. As the noise level increases, the classification accuracy of all methods decreases, indicating progressively weaker separation among the pathway profiles.

Table 4. Pathway-classification accuracy under increasing indicator noise and corresponding ASI values (mean of 10 random seeds)

Noise	S_6	S_4	Euclid.	Cosine	5-NN	Logistic	Forest	SVM	LDA $\rightarrow S_6$	ASI S_6	ASI LDA $\rightarrow S_6$
0.5	98.5	98.3	99.1	99.1	98.8	98.9	98.5	98.9	98.7	0.602	0.815
1.0	88.3	86.4	90.3	90.2	87.4	89.8	88.4	89.3	88.4	0.306	0.547
1.5	76.9	74.2	80.4	79.6	74.9	79.9	77.3	78.5	78.4	0.189	0.422
2.0	68.1	64.9	72.0	71.2	64.8	71.6	68.8	70.3	70.5	0.135	0.352

Note: Accuracy values are percentages. S_6 and S_4 denote the corresponding fuzzy prototype matchers. LDA $\rightarrow S_6$ denotes two-dimensional linear discriminant analysis followed by fuzzification and S_6 -based prototype matching. 5-NN = five-nearest neighbors; SVM = support vector machine; ASI = Aggregate Separation Index.

At the lowest noise level of 0.5, all methods achieve accuracies close to 99%. The best crisp prototype accuracy is 99.1%, attained jointly by the Euclidean and cosine prototype classifiers, while the S_6 matcher achieves 98.5%. At a noise level of 1.0, the Euclidean prototype reaches 90.3%, compared with 88.3% for S_6 . The corresponding accuracies at noise levels of 1.5 and 2.0 are 80.4% versus 76.9%, and 72.0% versus 68.1%, respectively.

Across the four tested noise levels, the S_6 prototype matcher therefore remains 0.6–3.9 percentage points below the best crisp prototype benchmark. Figure 7a shows that this performance gap generally widens as the indicator noise increases. The difference is only 0.6 percentage points at the lowest noise level but reaches 3.9 percentage points under the highest tested noise.

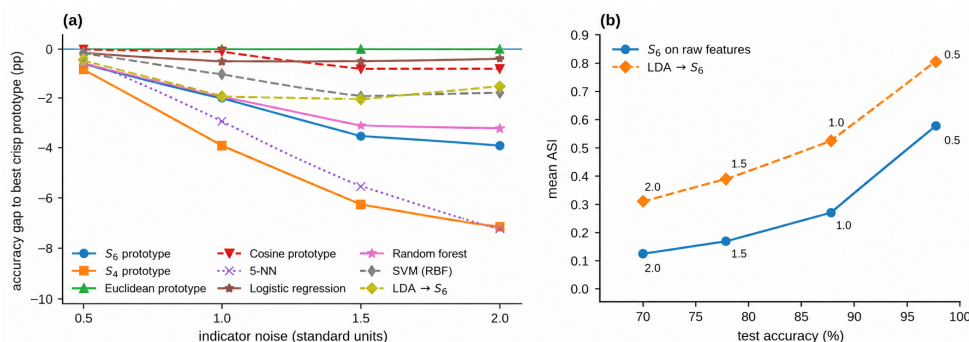


Figure 7. Machine-learning benchmark results: (a) accuracy gaps relative to the best crisp prototype under increasing indicator noise; (b) ASI versus test accuracy for S_6 matching with raw features and LDA embedding

Note: 5-NN = five-nearest neighbors; SVM (RBF) = support vector machine with radial basis function kernel; ASI = Aggregate Separation Index; LDA = linear discriminant analysis.

The result indicates that the uncertainty-aware similarity representation introduces only a moderate predictive penalty relative to the strongest crisp prototype classifier. This comparison is particularly relevant because the proposed framework is not designed solely to maximize classification accuracy. In addition to assigning an observation to a reference pathway, it preserves information on similarity to competing prototypes and provides an interpretable measure of separation among the resulting matches.

The benchmark should therefore be interpreted as an assessment of the predictive cost associated with retaining an uncertainty-aware and interpretable decision representation. The results show that this additional decision information is obtained without a substantial loss of classification performance across the tested noise conditions.

6.4 Effect of the Linear Discriminant Analysis Representation

A two-dimensional LDA embedding [30] is introduced before the fuzzification stage to examine whether a more compact discriminative representation can improve the performance of the similarity-based matcher.

Let the original normalized feature vector be $x \in \mathbb{R}^d$. The LDA transformation maps this vector into a lower-dimensional representation, $z = W^\top x$, where $z \in \mathbb{R}^2$.

The embedded representation is subsequently fuzzified and evaluated using the same similarity-based reasoning structure. At the two highest noise levels, the LDA-assisted matcher reduces the performance gap to approximately 1.5–2.0 percentage points at the two highest noise levels. The dimensionality reduction also increases the mean ASI by approximately 1.4–2.6 times, depending on the noise level. This result suggests that a compact discriminative representation can improve both class separation and the ASI produced by the fuzzy matching layer. However, LDA should be interpreted as a preprocessing benchmark rather than as a required component of the proposed framework.

6.5 Relationship Between Aggregate Separation and Classification Accuracy

The benchmark also examines whether the ASI responds meaningfully as the classification problem becomes more difficult. For the similarity-based matcher, the mean ASI decreases consistently as test accuracy declines.

At the lowest noise level, the S_6 matcher achieves an accuracy of approximately 98.5%, with a mean ASI of about 0.60. At the highest tested noise level, the accuracy falls to approximately 68.1%, while the mean ASI decreases to about 0.14. This indicates that weaker classification performance is accompanied by reduced separation among the competing prototype matches.

As shown in Figure 7b, the same overall pattern is observed for both the raw-feature and LDA-assisted matchers. The LDA representation maintains higher ASI values across the tested noise levels, indicating stronger separation among the competing prototype matches after dimensionality reduction.

This behavior is useful in practical operation because ground-truth labels may not be available at the time a decision is made. In such situations, a lower ASI can provide an additional indication that the current classification is less distinctly separated from competing alternatives and should therefore be interpreted with greater caution.

The ASI should not be interpreted as a probability or as a calibrated measure of predictive confidence. Rather, it quantifies the aggregate separation between the leading match and the competing alternatives. Its decline under increasing noise therefore provides complementary information on the distinctiveness of the resulting decision.

6.6 Interpretability Relative to Conventional Classifiers

The principal advantage of the similarity-based layer is not that it outperforms every conventional classifier in predictive accuracy.

Rather, it provides additional decision information that is directly interpretable.

For each observation, the framework can report:

- similarity to each reference profile;
- the fuzzy-information representation of the indicators;
- hesitation or abstention where applicable;
- the separation between competing alternatives; and
- the ASI associated with the leading match.

A conventional classifier such as a random forest can provide a predicted class and a voting or probability-related output, but it does not naturally express the decision in terms of the original energy-planning criteria.

For a planning or supervisory energy-management application, this distinction is important. Decision makers may need to know not only which pathway or operating state is preferred, but also how closely the current condition resembles alternative profiles and whether the resulting recommendation is sufficiently distinct to support action.

6.7 Implications for Smart-Energy Decision Support

The benchmark supports the use of the proposed fuzzy similarity layer as an interpretable component within a broader smart-energy architecture. The additional uncertainty representation and separation information are obtained with only a modest reduction in classification performance relative to the best crisp prototype benchmark, making the approach suitable for applications in which decisions must remain transparent and auditable.

In practical implementation, predictive models could estimate waste composition, recovered-energy availability, electricity and heat demand, market prices, or equipment states. These outputs could then enter the uncertainty-aware decision layer, where competing alternatives are evaluated using the same weighting, similarity, and robustness mechanisms applied in the preceding case studies.

This separation of functions allows machine learning to support data processing and forecasting without becoming an opaque final decision rule. The predictive component supplies updated system information, while the fuzzy decision layer preserves criterion-level interpretability and indicates how clearly the leading alternative is separated from competing options.

7 Discussion

7.1 Implications for Sustainable Waste-to-Energy Planning

The results show that sustainable WtE planning should be treated as a sequence of connected resource, infrastructure, and operational decisions rather than as a single technology-selection problem. The cases examined in this study span waste-valorization pathway selection, district-to-pathway matching, facility siting, conversion-route selection, plant operation, and industrial-symbiosis partner selection. This broader perspective is important because the performance of a WtE system depends not only on the selected conversion technology, but also on feedstock characteristics, infrastructure access, operating conditions, and the downstream utilization of recovered energy and by-products.

The pathway-selection results illustrate this point clearly. AD ranks first under the present scenario, but its leading position reflects the combined influence of energy recovery, greenhouse-gas reduction, cost, land requirement, social acceptance, and regional maturity rather than energy output alone. Recent studies similarly show that WtE pathway selection requires simultaneous consideration of technical, environmental, economic, and broader sustainability criteria [31–35]. A technology with higher gross energy production may therefore remain less attractive if it performs poorly in terms of carbon intensity, infrastructure requirements, cost, social acceptance, or local implementation conditions.

The district-matching and siting results further demonstrate the importance of spatial context. Waste composition affects the suitability of biological, thermal, and material-recovery pathways, while facility value depends partly on access to the grid and suitable users of recovered heat. Recent WtE siting research has likewise emphasized the role of public participation and broader social and environmental considerations alongside technical and economic suitability [36]. For CHP-based systems, the practical value of recovered heat also depends on the location and demand characteristics of potential users. Studies of industrial excess-heat recovery and district-heating integration show that spatial proximity, network requirements, heat-demand profiles, and economic conditions can strongly influence the feasibility of industrial energy symbiosis [37–39].

7.2 Energy-Conversion and Operational Trade-Offs

The conversion-route case provides a direct comparison of alternative energy pathways. Biogas CHP ranks first under all six similarity kernels, followed by biomethane injection and WtE-CHP. This result reflects the combined value of electricity and useful heat production together with dispatchable operation, but it should not be interpreted as evidence that CHP is universally preferable to biomethane, WtE-CHP, RDF co-firing, or hydrogen production. The preferred route depends on the criterion structure and on local opportunities for heat utilization, grid interaction, fuel substitution, and infrastructure integration. For example, biomethane may become more attractive where suitable organic feedstocks, gas-upgrading and injection infrastructure, and long-term gas demand are available, consistent with recent assessments of biowaste-to-biomethane pathways [40]. WtE-CHP, by contrast, may perform more favorably where stable electricity and thermal offtake can be maintained.

The operating-strategy case highlights a different trade-off. Under entropy-derived weights, Baseload operation is preferred because forecast sensitivity, equipment wear, and operational complexity contribute strongly to the ranking. As greater importance is assigned to revenue and grid-stability contribution, Hybrid-adaptive operation becomes preferable. The ranking reversal shows that the preferred CHP strategy depends on whether the plant is expected primarily to provide stable energy output or to participate more actively in market-responsive and grid-support operation. This distinction is particularly relevant to smart-energy systems in which dispatchable distributed resources are increasingly expected to respond to changing prices, grid conditions, and thermal demand.

7.3 Importance of Uncertainty and Decision Separation

A central finding of the study is that uncertainty affects not only the ranking of alternatives, but also the degree to which the leading option is separated from its competitors. This is evident in the conversion-route case, where removing the stakeholder-abstention component does not change the ordering of the leading technologies but shifts the ranking indices by as much as 0.020 and the ASI values by as much as 0.026. The same first-ranked technology can therefore emerge under different uncertainty representations while the strength of its separation from competing alternatives changes.

The industrial-symbiosis case provides an even clearer example. Under the reference measure, the cement plant and food processor obtain ranking indices of 0.613 and 0.605, respectively, leaving an adjacent gap of only 0.008. The identity of the leader also changes across similarity kernels, while the Monte Carlo probability that the nominal

leader remains first is only 0.632. Under these conditions, presenting the cement plant as a definitive best partner would overstate the strength of the available evidence.

The framework therefore identifies the decision as contested rather than forcing a unique recommendation. In practical terms, such a result suggests that collecting better information may be more valuable than further refinement of the ranking algorithm. For the partner-selection case, improved evidence on long-term offtake capacity, contractual commitment, or actual integration requirements would provide a stronger basis for distinguishing the leading alternatives.

7.4 Robustness of the Principal Recommendations

The Monte Carlo analysis distinguishes clearly between robust leading alternatives and less stable lower-order rankings. Under the reference perturbation setting, the pathway-selection leader remains first with a probability of 0.948, while the corresponding probability for the CHP operating-strategy case reaches 0.999. These results indicate that the principal recommendations in both cases are highly resistant to the specified level of rating perturbation.

Full-order stability is considerably lower, ranging from approximately 0.24 to 0.72, which indicates that lower-ranked alternatives can exchange nearby positions even when the leading alternative remains unchanged. Nevertheless, the mean Spearman rank correlation remains above 0.93, showing that the overall ranking structure is largely preserved. The robustness assessment should therefore consider both leader-survival probability and full-order stability rather than relying on a single indicator.

7.5 Criterion Weights Are More Influential Than the Similarity Kernel

Across the case studies, changes in criterion weights have a stronger effect on the principal decisions than changes in the admissible similarity kernel. In the pathway case, replacing entropy-derived weights with equal weights moves material recovery ahead of CO while AD remains first. In the CHP operating-strategy case, increasing the importance of revenue and grid contribution eventually changes the preferred strategy from Baseload to Hybrid-adaptive. By contrast, variation among the valid similarity kernels has relatively little effect on the leading alternatives.

This finding indicates that decision outcomes are driven more strongly by the priorities assigned to the evaluation criteria than by the specific similarity operator. Energy, economic, environmental, and operational priorities should therefore be defined carefully, because they determine how competing objectives are balanced. The role of the proposed framework is not to eliminate these trade-offs, but to make their influence on the final decision explicit.

7.6 Value of the Hybrid Fuzzy Representation

The case studies show that different forms of uncertainty are better represented by different fuzzy-information structures. The pathway decision contains membership and non-membership combinations that fall outside the intuitionistic domain but remain admissible within the Pythagorean setting. The CHP operating-strategy case requires a Fermatean representation, whereas the conversion-route case includes an explicit abstention component that is retained through the T -spherical formulation.

This flexibility avoids forcing heterogeneous information into a single uncertainty model. In energy planning, uncertainty may arise from incomplete technical information, stakeholder hesitation or neutrality, or uncertain commercial conditions. The hybrid representation allows these different structures to be handled within the same decision framework while preserving information that would otherwise be lost. The choice of fuzzy family should therefore be determined by the structure of the underlying data rather than by methodological complexity alone.

7.7 Role of the Similarity Measure

The comparison of the six extended sine-based kernels shows that not all mathematically valid similarity functions are equally suitable for unrestricted decision analysis. The restricted monotonicity of S_1 is particularly important because a similarity measure used in energy planning should not assign a larger similarity to an alternative that moves farther away from the ideal energy-performance profile.

Measures satisfying the complete set of required properties are therefore more appropriate for the principal analysis, while kernels with restricted monotonicity or metric behavior are better retained for comparison and sensitivity testing. Among the tested measures, S_6 provides a parameter-free option with the required mathematical properties and stable case-study results. At the same time, the main energy conclusions remain largely consistent across the admissible kernels, indicating that the practical recommendations are not artifacts of a single similarity function.

7.8 Machine Learning and Interpretability

The machine-learning benchmark highlights the trade-off between predictive performance and interpretability. Across the tested noise levels, the fuzzy similarity matcher remains within approximately 0.6–3.9 percentage points of the best crisp prototype benchmark while retaining similarity information for multiple reference profiles, hesitation

information, and the ASI. The mean ASI also decreases as the classification problem becomes more difficult, falling from approximately 0.60 at high classification accuracy to about 0.14 under the highest tested noise. This behavior indicates that weaker predictive performance is accompanied by reduced separation among competing prototype matches.

The proposed framework should therefore be viewed as complementary to, rather than a replacement for, machine learning. Predictive models can process large operational datasets or generate forecasts, while the uncertainty-aware similarity layer provides a transparent basis for strategic and supervisory decisions. Such an arrangement is particularly relevant to smart-energy applications in which model outputs may need to be interpreted by plant operators, utilities, municipalities, or other stakeholders before investment or operating decisions are made.

7.9 Practical Implementation

The five-layer framework can support decisions operating at different temporal scales. Strategic choices such as conversion-technology selection, facility siting, and industrial-partner selection would normally be reconsidered only when substantial new information becomes available, whereas operational decisions, particularly CHP operating strategy, could be updated more frequently in response to changes in electricity prices, thermal demand, grid requirements, or forecast uncertainty. In a practical deployment, updated information from SCADA platforms, smart meters, municipal databases, and energy-market or forecasting systems could be used to revise the decision matrix and recalculate criterion weights, alternative similarities, ASI values, and rank-stability indicators.

The resulting framework should be interpreted as a supervisory decision-support layer rather than as a replacement for detailed physical models or low-level plant control. Its role is to provide a transparent structure for comparing strategic and operational alternatives under heterogeneous and uncertain information. Detailed mass and energy balances, equipment dynamics, dispatch optimization, and market-clearing processes are outside the scope of the present study, but their outputs could be incorporated as inputs to the decision layer in future implementations.

7.10 Limitations

Several limitations should be considered when interpreting the results. Most importantly, the case-study data are illustrative rather than measurements from an operating WtE facility. The numerical results are therefore intended to demonstrate the behavior and robustness of the decision framework and should not be interpreted as empirical evidence that a particular technology, site, operating strategy, or industrial partner is universally preferable. In addition, each decision is represented through a single decision matrix, whereas practical infrastructure planning may involve assessments from multiple experts, stakeholder groups, utilities, municipalities, and industrial partners.

The weighting and uncertainty treatments also have limitations. Entropy weighting reflects the information dispersion contained in the decision matrix but does not necessarily represent policy or strategic importance. A criterion may receive a relatively small entropy weight because the alternatives perform similarly on that criterion even when decision makers consider it highly important. The expert-weight analysis in Case 3 demonstrates that subjective priorities can materially affect the preferred energy strategy. Likewise, the Monte Carlo analysis perturbs the existing evaluations but does not distinguish explicitly among measurement error, forecast uncertainty, expert disagreement, parameter uncertainty, and scenario uncertainty.

The machine-learning benchmark is based on synthetic profiles rather than a large operational dataset, so its purpose is limited to examining the predictive cost and separation behavior of the similarity-based matcher. Validation with real SCADA, waste-composition, energy-demand, and operating data would be required before drawing conclusions about industrial predictive performance. The current framework also does not explicitly model detailed life-cycle emissions, physical mass and energy balances, or dynamic grid constraints. These elements could be incorporated in future applications by linking the decision layer with life-cycle assessment, process simulation, or energy-system optimization models.

8 Conclusion

This study developed an uncertainty-aware intelligent decision-support framework for sustainable WtE planning and smart-energy infrastructure. The framework extends sine-based similarity measures from intuitionistic fuzzy information to q -rung orthopair and T -spherical fuzzy environments and integrates them with entropy-based weighting, ideal-solution comparison, the ASI, and Monte Carlo rank-stability analysis. The framework was examined through four illustrative cases covering waste-valorization pathway selection, facility siting, energy-conversion route selection, operating-strategy selection for a biogas CHP unit, and industrial-symbiosis partner selection.

The results show that the framework can distinguish between robust and contested decisions rather than simply returning a unique ranking in every case. AD is consistently identified as the preferred waste-valorization strategy under the present decision matrix, while biogas CHP ranks first among the tested conversion routes. Under entropy-derived weights, Baseload is the preferred CHP operating strategy, although Hybrid-adaptive operation becomes more attractive when revenue and grid-support objectives receive greater importance. The pathway and operating-strategy

decisions exhibit high rank-1 stability under the reference perturbation setting, with leader-survival probabilities of 0.948 and 0.999, respectively. By contrast, the industrial-symbiosis partner decision remains unresolved because the two leading candidates differ by only 0.008 and the nominal leader retains first place with a probability of only 0.632.

The analysis further shows that criterion weighting has a stronger influence on the principal decisions than the choice among the admissible similarity kernels. The machine-learning benchmark also indicates that the similarity-based matcher remains within 0.6–3.9 percentage points of the best crisp prototype benchmark across the tested noise levels while retaining explicit information on similarity, hesitation, and aggregate separation. These results support the use of the proposed framework as an interpretable supervisory decision layer within a broader smart-energy architecture.

The main limitation is that the case-study data are illustrative rather than derived from a fully operating WtE facility; the reported rankings should therefore be interpreted as demonstrations of the framework rather than universal technology recommendations. Future work should validate the method using operational and municipal data, incorporate multiple experts and stakeholder groups, improve the integration of objective and subjective criterion weights, and distinguish among different sources of uncertainty. Further extensions could also connect the decision layer with life-cycle assessment, process simulation, detailed mass and energy balances, and dynamic energy-system optimization. Overall, the framework provides a transparent approach for linking waste valorization, energy recovery, conversion technology, plant operation, and industrial symbiosis within a common uncertainty-aware decision structure.

Author Contributions

Conceptualization, M.S. and J.M.M.; methodology, M.S.; software, M.S.; validation, M.S. and J.M.M.; formal analysis, M.S.; investigation, M.S.; resources, M.S.; data curation, M.S.; writing—original draft preparation, M.S. and J.M.M.; writing—review and editing, M.S.; visualization, M.S.; supervision, M.S. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

Declaration on the Use of Generative AI and AI-assisted Technologies

During the preparation of this work, the authors utilized ChatGPT (OpenAI) for language polishing and improving readability. Afterward, they reviewed and edited the content as necessary and take full responsibility for the publication's content.

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